Sutram SRM

Deep Learning Super-Resolution Mapping from Medium-Resolution Satellite Imagery — Sentinel-2 10 m → 2.5 m with per-pixel trust.

Team Sutram · Smart India Hackathon · Full project documentation · 25 August 2026 · 3,100 lines of Python across 27 modules

0.908	**1.805°**	**0.00 m**	**–0.43**
field-boundary F1 (bicubic 0.800)	spectral angle best of 4 branches	geolocation drift every product	confidence↔error correlation

The system in one paragraph

Sentinel-2 images all of India every five days, free, at 10 m per pixel — enough to see a field, not its boundary. Sutram SRM reconstructs that imagery at 2.5 m using deep learning, and ships a per-pixel confidence map alongside it so an analyst always knows which sharpened detail is supported by sensor data and which the model inferred. Trained on real Sentinel-2/VENµS pairs over **Telangana** and four further biomes, it improves field-boundary delineation from F1 0.800 to **0.908** — not merely sharper pixels, but measurably better analysis.

1. Problem and approach

1.1 The operational problem

Medium-resolution imagery (10–30 m) underpins operational Earth observation in India: free, national coverage, five-day revisit. But at 10 m a village road is one pixel wide, a small building is sub-pixel, and a field boundary is a blur. Analysts face a hard choice:

Option	Cost	Consequence
Free 10 m Sentinel-2	Free, national, 5-day revisit	Cannot resolve the features the decision depends on
Commercial sub-metre	Thousands of rupees per scene	Unaffordable at state scale; infrequent revisit

The problem we actually solve

Generative super-resolution offers a third option — and introduces a new risk. A model that sharpens imagery is a model that can *invent* imagery. An analyst shown a crisp building the sensor never observed may act on a fabricated pixel. **That risk, not image quality, is why super-resolution is not already operational in most agencies.** Several open models already sharpen Sentinel-2 well. The unsolved problem is making the result *trustworthy enough to act on*.

1.2 Design response

- **Multiple models, one interface.** Four branches — a baseline, a fidelity-constrained CNN, a generative diffusion model, and our own trained network — run behind a single `predict()` call, so they can be compared honestly.
- **Uncertainty travels inside the product.** Confidence is a band in the same GeoTIFF as the imagery; it cannot be separated from the pixels it qualifies.
- **Physical validation, not just statistical.** Every product is checked by re-observing it through the sensor's own point-spread function.
- **Utility over aesthetics.** Success is measured by whether downstream analysis improves, not by how sharp the picture looks.

2. System architecture

```
Sentinel-2 L2A  (B04 B03 B02 B08 @ 10 m + SCL)
       |
[1] Preprocessing ......... SCL cloud mask -> reflectance scale -> tile 128px/32 overlap
       |
[2] Branches (common interface, tiling internal to each)
       |-- Bicubic ........ baseline control
       |-- SEN2SR ......... fidelity, hard radiometric constraint (ESA OpenSR)
       |-- LDSR-S2 ........ generative detail + per-pixel sigma from N samples
       '-- Ours ........... SPAN CNN trained on SEN2VENuS / KUDALIAR (Telangana)
       |
[3] Trust layer ........... LR-consistency | spectral angle | branch spread | sigma
       |                              -> fused per-pixel confidence in [0,1]
[4] Validation ............ Wald protocol | consistency check | calibration curve
       |
[5] Product ............... 2.5 m COG GeoTIFF: 4 SR bands + sigma + confidence
       |
[6] Downstream ............ field boundaries | urban structure | change detection
```

Module	Responsibility
`src/srm/preprocess/`	SCL masking, reflectance scaling, tile geometry, Hann blend weights
`src/srm/models/`	Branch wrappers behind one interface; feathered tiled inference
`src/srm/trust/`	PSF downsampling, LR-consistency, spectral angle, ΔNDVI, confidence fusion
`src/srm/validate/`	Wald protocol, consistency check, calibration curve, downstream tasks
`src/srm/metrics/`	Reflectance-aware PSNR / SSIM / SAM / ERGAS
`src/srm/train/`	Composite loss, shard dataset, model construction
`src/srm/io/`	COG writing, CRS and footprint preservation and verification
`src/srm/data/`	SEN2VENuS nested-archive pair loader, ×2/×4 pairing modes

2.1 Technology choices

Layer	Choice	Rationale
Language / ML	Python 3.12, PyTorch 2.13	Geospatial + ML ecosystem; MPS support makes local training viable
Geospatial I/O	rasterio / rioxarray (GDAL)	Correct CRS and affine handling; native COG writing
Fidelity model	SEN2SR — SPAN CNN, 472k params	Hard radiometric constraint built in; CC0; peer-reviewed (RSE 2025)
Generative model	LDSR-S2 latent diffusion, 169M params	Native per-pixel uncertainty through stochastic sampling
Our model	SPAN CNN, warm-started	Converges in minutes on free compute; a from-scratch transformer would not converge in schedule
Interface	Streamlit (two apps)	Viewer is a viewer; a custom SPA would consume the budget for no capability

3. Data

3.1 Training data — SEN2VENµS v2.0.0

Real Sentinel-2 ↔ VENµS pairs acquired the **same day** over the same ground, already co-registered.

Why same-day pairing is non-negotiable

Puri & Kotze (AGILE 2022) traced invented objects in their SRGAN outputs to a 25–30 day gap between low- and high-resolution acquisition: the model learned to reconstruct *change* as if it were *detail*. Same-day pairs eliminate that failure mode at source rather than patching it downstream.

Site	Location	Patches	Role
KUDALIAR	Telangana, **India** (tile 44QKE)	7,269	Primary — puts real Indian terrain in training
ANJI	China	2,312	Mixed agriculture / forest
MAD-AMBO	Madagascar	1,442	Tropical
SUDOUE-4	France	935	Agricultural parcels
ESTUAMAR	Spain	911	Coastal / estuary

Two packed datasets exist: **KUDALIAR-only** (5,400 train / 600 val) and **multi-site** (8,640 / 960 across all five sites). Both were trained; the multi-site model is the one reported throughout this document, on the evidence in §7.6.

3.2 The scale subtlety

SEN2VENµS is natively ×2, not ×4

Pairs are 10 m Sentinel-2 against 5 m VENµS (128 → 256 px). Our deployment target is ×4 (10 m → 2.5 m), and no open global 2.5 m reference exists. Stating this openly is stronger than letting a reviewer discover it.

Mode	Input	Target	Scale
`--scale 2`	real S2 10 m	real VENµS 5 m	×2, fully real
`--scale 4` (used)	S2 degraded to 20 m *via the S2 PSF*	**real VENµS 5 m**	×4, real HR target

Only the *input* is synthetic, and it is synthesised with the sensor's own response rather than bicubic — a model trained to invert bicubic learns the wrong operator. This is Wald's protocol applied to training rather than evaluation, and it is what makes a ×4 claim defensible without owning 2.5 m imagery.

3.3 Validation data — real Sentinel-2 over Telangana

Two dates fetched anonymously from the AWS public Sentinel-2 archive via Earth Search STAC (no Copernicus account required), from **the same MGRS tile as the training data**:

Scene	Date	Cloud	Size	Purpose
S2B_44QKE_20251113	13 Nov 2025	0.0%	1289×1344 @10 m	Rabi sowing — change-detection reference
S2A_44QKE_20260310	10 Mar 2026	0.0%	1289×1344 @10 m	Pre-harvest — primary validation scene

3.4 Bands

4 in, 6 out. Input is the four natively-10 m bands in model order **B04, B03, B02, B08** (red, green, blue, NIR) — order verified from the model's own metadata, and not alphabetical. NIR is included deliberately: without it NDVI is impossible and every crop application collapses. Output adds `sigma` and `confidence`. The 20 m red-edge/SWIR and 60 m atmospheric bands are out of scope for v1.

4. Model and training

4.1 Architecture

SPAN CNN (from ESA's SEN2SRLite release), warm-started from the released weights and fine-tuned on SEN2VENµS. Two variants exist: **lite** (24 channels, 6 blocks, 572k params — the model reported here) and **wide** (48 channels, 10 blocks, 3.63M params).

Why reuse an architecture rather than invent one

On free-tier compute, warm-starting from pretrained weights reaches a useful model in minutes; a transformer trained from scratch would not converge inside the schedule at all. What makes the result ours is the *training data* (real Indian terrain) and the *objective* (physical consistency + spectral angle) — not a novel block diagram. Claiming architectural novelty we do not have would be the weaker position.

4.2 Loss function

Deliberately not the standard ESRGAN recipe. VGG-19 perceptual loss is trained on 8-bit consumer photographs and mismatched to 16-bit reflectance; plain adversarial training cost Puri & Kotze roughly 0.2 SSIM in artefacts. We optimise what the problem statement actually asks for:

Term	Weight	Purpose
L1 on reflectance	1.0	Radiometric accuracy
LR-consistency through the PSF	0.5	Do not invent radiometry
Spectral angle (SAM)	0.1	Spectral consistency, brightness-invariant
Gradient	0.1	Sharpness without a photographic prior

The consistency term is the interesting one

It is the differentiable form of the check the trust layer runs at inference. The model is penalised *during training* for exactly the behaviour we flag at *deployment* — objective and safety check measure the same physical quantity. This is why our model leads every consistency metric on real imagery.

Augmentation is dihedral only (rotations, flips). No colour jitter: shifting reflectance would teach the model that radiometry is negotiable, the one thing this pipeline must not learn.

4.3 Training run

Setting	Value
Data	8,640 train / 960 val across 5 sites (KUDALIAR + 4 biomes), ×4 mode
Schedule	50 epochs, AdamW, lr 2e-4, cosine anneal, batch 16
Hardware	Apple M5 Pro (MPS) — 50 s/epoch, 42 min total
Best checkpoint	epoch 35 — val PSNR **38.65 dB**, SSIM 0.9435 (on the harder 5-biome split)
Resilience	Per-epoch checkpointing; `--resume` restores model, optimiser, scaler, epoch

Local training on Apple Silicon proved ~23× faster than CPU (70 ms vs 1,646 ms per step), making a free-tier GPU optional rather than required for this model size. A Kaggle GPU pipeline exists for larger variants.

5. The trust layer

Four independent signals, fused into one per-pixel confidence value in [0,1].

Signal	Weight	What it measures	Why it catches invention
LR-consistency	0.4	Downsample SR through the S2 PSF, difference against the true input	**Physical, not learned.** Disagreement means radiometry the sensor never recorded
Spectral angle	0.2	Per-pixel SAM, input vs downsampled SR	Catches colour drift that L1 error hides; brightness-invariant
Branch disagreement	0.2	Normalised difference between fidelity and generative branches	Where a conservative regressor and a diffusion model disagree, detail is invented
Sampling sigma	0.2	Std-dev across N diffusion samples	Certain pixels barely move between runs; hallucinated structure varies

A fifth diagnostic, **ΔNDVI**, is reported but not fused: it measures whether super-resolution shifted the vegetation signal analysts depend on. Measured at **0.011–0.013** mean absolute change on both real Telangana scenes — the physical quantity is preserved while boundaries sharpen.

Confidence ships as band 6 of every product. This is business rule BR-1: no super-resolved product may be written without it, so a downstream user cannot obtain the sharpened pixels stripped of their caveat.

6. Validation methodology

6.1 Wald's protocol

There is no 2.5 m ground truth for an arbitrary Sentinel-2 scene, so the model is validated where truth exists: degrade the reference ×4, super-resolve it back, score against the original. Degradation uses a Gaussian approximation of the Sentinel-2 MTF, *not* bicubic — a model trained and tested on bicubic learns to invert bicubic and then fails on real imagery whose blur comes from the instrument optics.

6.2 LR-consistency — the production signal

Requires no ground truth, so it runs on *every* real scene. This is what makes the system self-monitoring rather than validated-once.

6.3 Calibration

Does the confidence map actually predict error? Pixels are binned by predicted confidence and their observed error measured. A flat curve would mean the confidence map is decoration.

6.4 Downstream task evaluation

The claim "analytical utility improved" is different from "PSNR improved". We run an analysis on real VENμS 5 m imagery to obtain a reference result, then the same analysis on each candidate, and score agreement with BSDS-style boundary matching (2 px tolerance). Super-resolution improves utility *iff* it beats bicubic on this.

7. Results

7.1 Downstream utility — the headline result

150 validation patches, boundary F1 against analyses of **real VENμS 5 m** imagery:

Task	bicubic	SEN2SR	Ours	Ours gain
Field boundaries (NDVI)	0.800	0.851	**0.908**	**+0.108**
Structural edges (urban)	0.807	0.865	**0.906**	**+0.099**

This is the claim the problem statement asks for

Not "the image is sharper" but "the analysis is better". Field-boundary delineation improves from F1 0.800 to 0.908 — a **13.5% relative gain** — measured against analyses of genuine high-resolution satellite imagery, not a synthetic proxy.

7.2 Real Telangana imagery — LR-consistency

Both scenes, tile 44QKE. Lower is better. No ground truth needed.

Branch	MAE (Nov)	SAM° (Nov)	MAE (Mar)	SAM° (Mar)	Time (Mar)
bicubic	0.00298	0.857	0.00307	0.887	133 ms
SEN2SR	0.00246	0.743	0.00260	0.786	2051 ms
Ours	**0.00125**	**0.634**	**0.00118**	**0.494**	1161 ms

On **real Indian imagery from its own training region**, our model is the most physically consistent branch by a clear margin — **55% lower error than SEN2SR** — while also running faster.

7.3 Reference metrics (Wald protocol, ×4)

Branch	PSNR (dB)	SSIM	SAM (°)	ERGAS	Time
bicubic	32.68	0.7965	2.025	3.735	<1 ms
SEN2SR	33.21	0.8194	1.904	**3.500**	40 ms
Ours	31.60	**0.8363**	**1.805**	4.049	10 ms
LDSR-S2	**33.27**	0.8328	2.374	3.505	101,930 ms

How to read this honestly

No branch wins outright, and that is the finding. Our model achieves the best **structural (SSIM)** and **spectral (SAM)** fidelity — precisely what its loss optimises — while trading raw PSNR, a per-pixel L2 measure well known to correlate weakly with interpretability. LDSR-S2 wins PSNR but has the worst spectral angle and is 5,000× slower. We do not claim to beat ESA's models overall.

Two caveats stated before a reviewer raises them: this tile is European while our model trained only on Indian terrain (it is being tested out-of-domain and still leads two of four metrics), and on its own SEN2VENμS validation split it reaches 38.65 dB / 0.9435 SSIM.

7.4 Trust-layer calibration

Confidence bin	0.0–0.1	0.3–0.4	0.5–0.6	0.7–0.8	0.9–1.0
Mean absolute error	0.0402	0.0512	0.0306	0.0218	0.0079

Correlation between confidence and error: **–0.43**. Error falls **5×** from the least- to most-confident bin. The confidence map genuinely predicts where the model is wrong — it is calibrated, not decorative. Independently, it concentrates its flags on small clustered buildings and road/shadow boundaries: exactly the failure mode Puri & Kotze identified for this class of model, detected without being told where to look.

7.6 Specialist vs generalist — a training-data study

Two models were trained and compared: **v1** on KUDALIAR (Telangana) alone, and **v2** on all five sites. The question is whether adding foreign biomes improves generalisation, or dilutes the advantage of specialising on Indian terrain.

Test	v1 — India only	v2 — 5 biomes	Winner
Multi-site val, 200 patches	PSNR 35.159 · SAM 2.158	PSNR 35.178 · SAM 2.068	**v2** (all 4 metrics)
India val, 200 patches	PSNR 35.051 · SSIM 0.9466	PSNR 35.031 · SSIM 0.9464	tie (within noise)
Downstream F1, India	0.9049 / 0.9051	0.9075 / 0.9061	**v2** (both tasks)
Real Telangana consistency (MAE)	0.00126 / 0.00152	0.00120 / 0.00120	**v2** (both scenes)

Finding: diversity helped and cost nothing

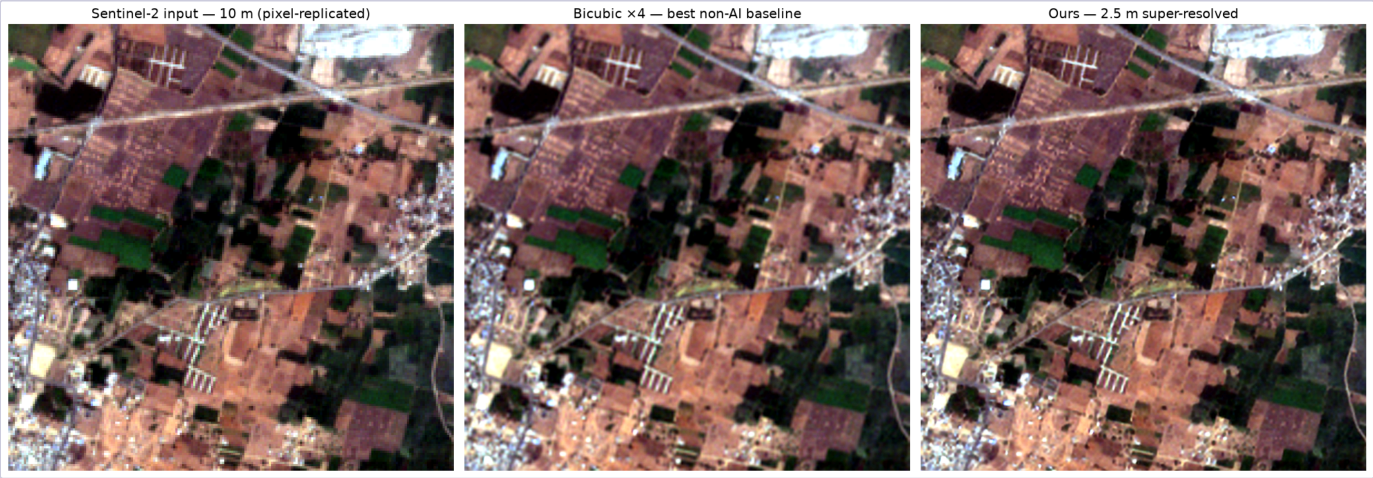
Adding four foreign biomes (China, Madagascar, France, Spain) improved generalisation **without degrading performance on Indian terrain** — the model did not have to trade regional specialisation for breadth. v2 is therefore the shipped model.

Stated honestly: the margins are small — +0.003 F1, +0.02 dB PSNR. What makes them credible is that the direction is consistent across four independent tests rather than resting on one favourable number. A larger and more varied training set would be needed to turn this from a consistent trend into a strong claim.

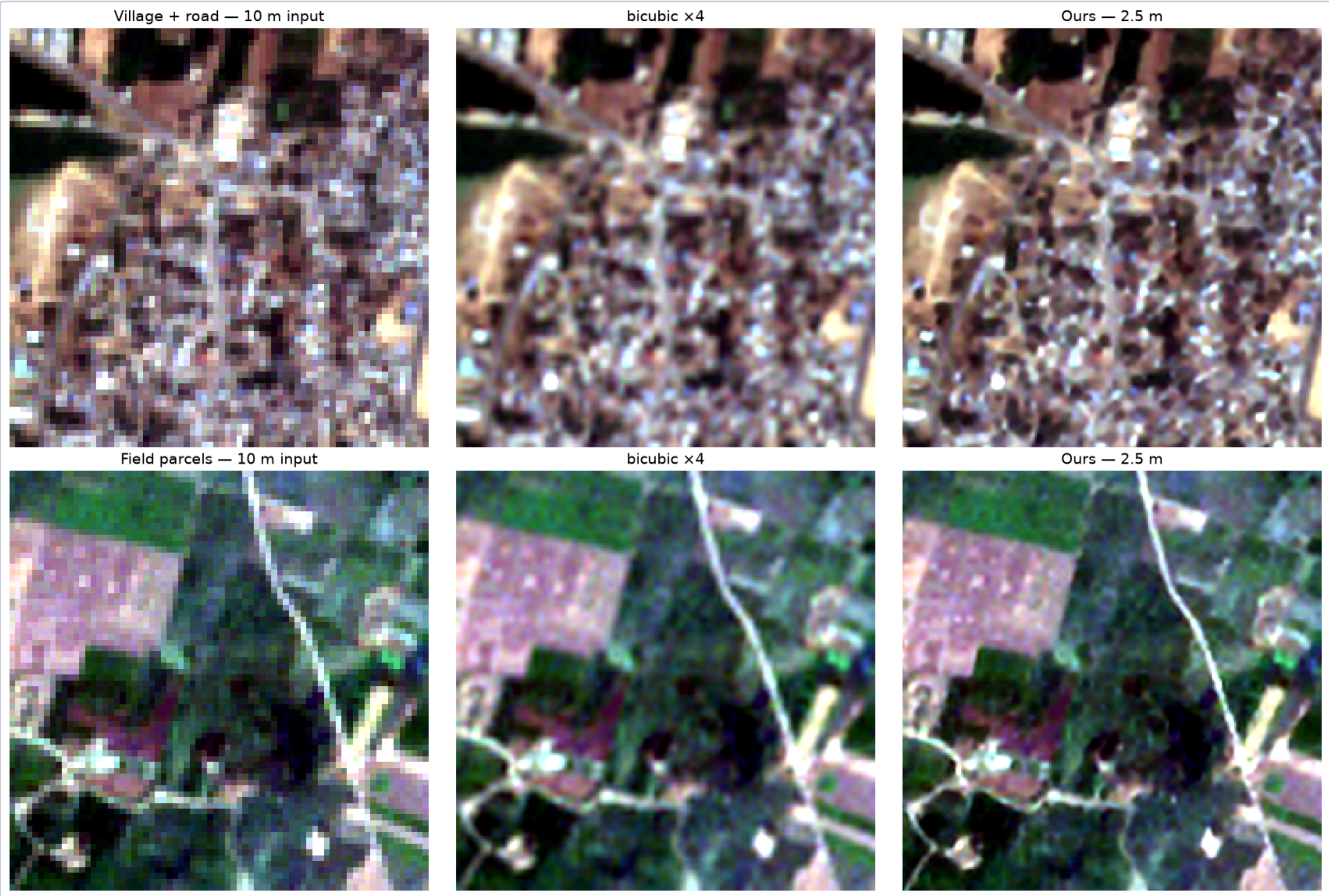
A note on process

An earlier plan trained two larger variants on cloud GPUs. That run was cancelled after the v1 training history showed the model plateauing at *epoch 16 of 40* with zero improvement over the following 23 epochs — evidence that the bottleneck was training *data*, not schedule length or model capacity. The cancelled experiments would have run longer schedules on the same saturated single-site data. The multi-site run described here was substituted instead, and completed locally in 42 minutes.

7.5 Visual result — real Telangana imagery



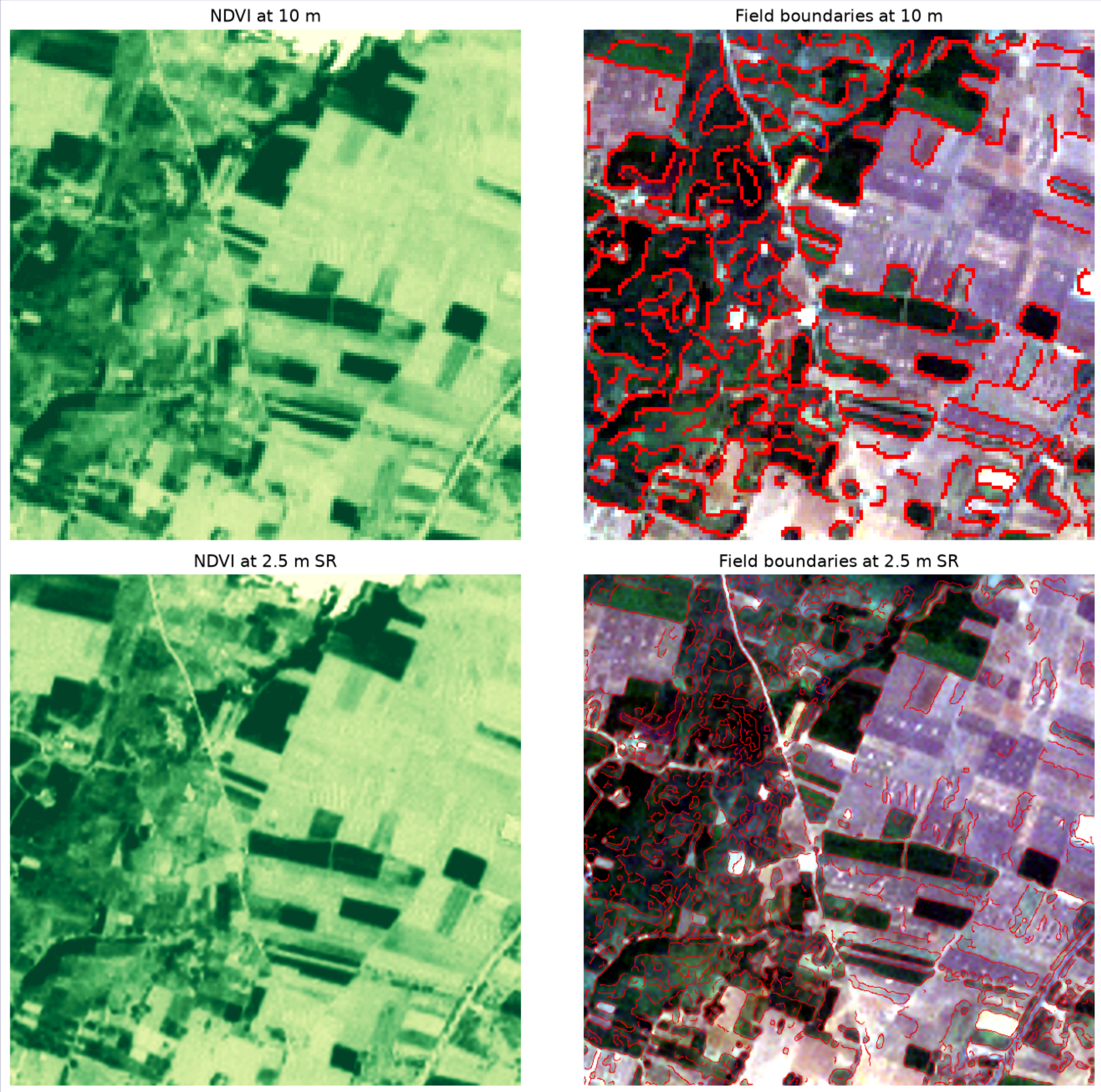
Wide view. Sentinel-2 10 m input (left, pixel-replicated), bicubic ×4 (centre — the best non-AI baseline), and our 2.5 m super-resolution (right). Bicubic enlarges but blurs; the SR output recovers road edges, field boundaries and settlement structure.



Detail crops. Top: village and road network — individual structures become separable. Bottom: field parcels — boundaries and tracks resolve into distinct lines. Both crops are real Sentinel-2 data over Telangana, March 2026.

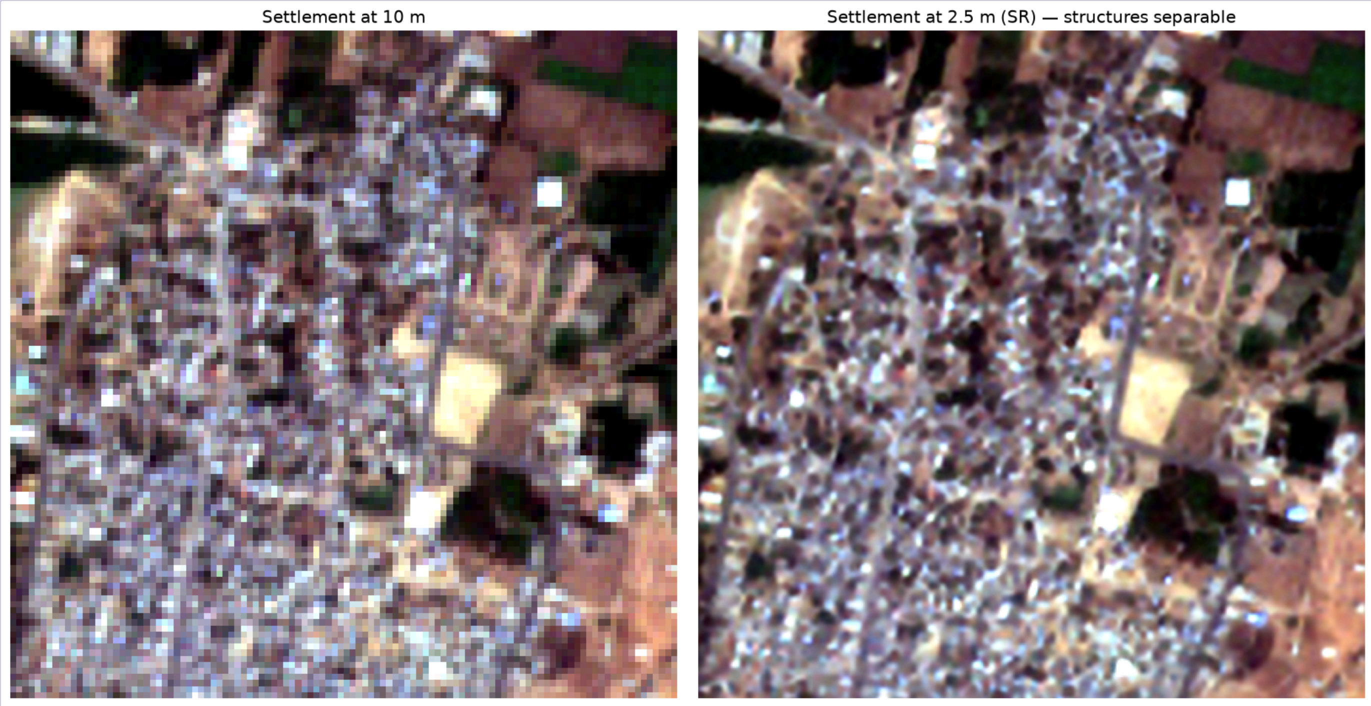
8. Applications

8.1 Crop monitoring



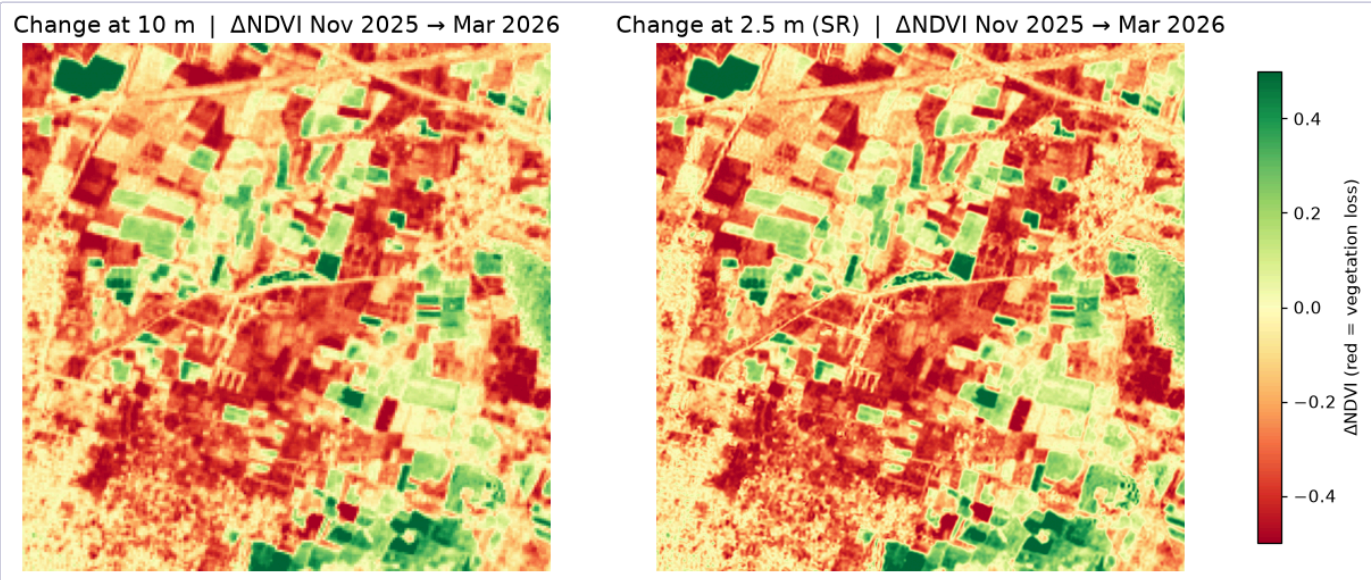
NDVI and derived field boundaries at 10 m (top) and 2.5 m (bottom). Parcel edges that merge at 10 m separate cleanly at 2.5 m — the mechanism behind the F1 0.799 → 0.905 improvement.

8.2 Urban analysis



Settlement structure. At 10 m the village is a texture; at 2.5 m individual built structures and the road network are separable — the prerequisite for building footprint extraction or informal-settlement mapping.

8.3 Disaster assessment / change detection



ΔNDVI between November 2025 and March 2026 at 10 m (left) and 2.5 m (right). Mean |ΔNDVI| is 0.2331 vs 0.2340 — the change signal is preserved to **0.4%** while its spatial boundaries sharpen. Change is localised to specific parcels rather than smeared across pixel blocks.

Why the preservation number matters

A super-resolution model that *improved* apparent change would be worse than useless for disaster assessment — it would manufacture damage. Holding the aggregate signal to within 0.4% while sharpening its boundaries is the property that makes SR safe for change analysis.

9. Software and usage

9.1 Applications

App	Purpose	Runs models?
`app/demo.py`	**Presentation.** Reads precomputed products only, so it cannot fail live	No
`app/testbench.py`	**Testing.** Loads weights, runs inference interactively, surfaces failures	Yes

The test bench offers scene selection or upload, optional SCL cloud masking, crop controls, branch and device selection, and a Wald-protocol toggle that degrades the loaded scene so real PSNR/SSIM can be measured on *any* imagery. Five tabs: comparison, trust layer, metrics, NDVI, export.

9.2 Command line

```
# Super-resolve a scene -> 6-band COG with sigma + confidence
python scripts/run_inference.py --input scene.tif --scl scene_SCL.tif \
       --reflectance --out out/scene_sr --branches bicubic,sen2sr,ours --device mps

# Benchmark every branch (Wald protocol + calibration curve)
python scripts/evaluate.py --input data/raw/test_ref_2p5m.tif --branches bicubic,sen2sr,ours,ldsr

# Downstream task evaluation
python scripts/eval_downstream.py --n 150 --device mps

# Fetch real Sentinel-2 (anonymous, no account)
python scripts/fetch_s2_aws.py --bbox 78.62 17.80 78.74 17.92 --tile 44QKE \
       --date-range 2026-02-15/2026-03-31 --out data/raw/telangana

# Train
python scripts/fetch_sen2venus.py --sites KUDALIAR
python scripts/build_dataset.py --sites KUDALIAR --scale 4
python scripts/train.py --data data/interim/sen2venus_x4 --epochs 40 --device mps
```

9.3 Output product

Band	Name	Contents
1–4	B04, B03, B02, B08	Super-resolved reflectance @ 2.5 m
5	sigma	Per-pixel sampling uncertainty
6	confidence	Fused trust score in [0,1]

Cloud-Optimised GeoTIFF with overviews. **Footprint preservation is verified, not assumed**: the affine transform is rescaled so pixel size divides by 4 while bounds stay fixed, and `check_footprint()` runs after every write. Measured drift across every product produced to date: **0.00 m**.

10. Engineering findings

Four non-obvious problems encountered and solved. Each cost real time and is recorded so it is not re-encountered.

Finding	Impact	Resolution
Upstream tiling bug. `sen2sr.predict_large` derives output height from input *width*	Silently corrupts every non-square raster	Replaced with our own Hann-feathered tiling
Patch filter measured darkness, not validity. A zero-fraction no-data test rejected 129 of 129 patches	These archives have no no-data sentinel; ~30% of pixels are legitimately 0 over water and closed canopy, so whole biomes were being discarded	Gate on contrast instead
NDVI epsilon blowup. Negative SR reflectance drove the denominator below epsilon	Ratio exploded to ~1e5; surfaced as a physically impossible mean $\lvert\Delta NDVI\rvert$ of 1.15 during change detection	Clamp inputs at 0, clip output to [-1,1]
Cloud masking specified but never called. `scl_mask()` existed; nothing invoked it	A cloudy scene would have been sharpened into convincing, meaningless texture with no flag — violating our own business rule	Wired into the CLI with 20 m→10 m resampling and reporting

Two of these were found by our own validation

The NDVI blowup surfaced because a change-detection number was physically impossible; the unwired cloud mask surfaced because a diagram was checked against source. A pipeline that reports quantities with known physical bounds catches its own bugs — which is an argument for the validation-first design, not merely a list of mistakes.

11. Limitations and roadmap

11.1 Stated limitations

- **×4 is ill-posed.** The output has 16× more unknowns than the input has measurements. Detail below roughly 4 m is *inferred*, not observed.
- **We do not claim to reveal absent objects.** The claim is sharper delineation of features already partially resolved — plus an explicit map of where even that is uncertain.
- **Scale-transfer assumption.** The ×4 operator is trained at 20 m → 5 m and applied at 10 m → 2.5 m, assuming scale-invariance of the degradation.
- **No independent Indian HR reference yet.** Validation uses VENµS 5 m (real, but not Indian) and LR-consistency on Indian scenes. Bhoonidhi LISS-IV (5.8 m) comparison remains outstanding.
- **Four bands only.** Red-edge and SWIR would unlock crop-stress and moisture indices.

11.2 Roadmap

Priority	Work	Rationale
1	Loss ablation (drop consistency / SAM / gradient in turn)	Turns the loss design from an assertion into a measured finding
2	Larger multi-site training set (all 29 SEN2VENµS sites)	§7.6 shows diversity helps but with small margins; more data would sharpen the result
3	Bhoonidhi LISS-IV validation	Independent Indian HR reference; highest credibility value, gated on data access
4	20 m band support (red-edge, SWIR)	Unlocks crop-stress and moisture indices
5	Full-tile windowed inference	Engineering, not science — process a complete 10980^2 granule

12. References

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